PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Information and Communication Technologies for the improvement of cognitive function in healthy older adults. A systematic review protocol
AUTHORS	Pastells-Peiró, Roland; Fernández-Lago, Helena; Rubinat Arnaldo, Esther; Bellon, Filip; Martínez-Soldevila, Jordi; Gea-Sánchez, Montserrat

VERSION 1 – REVIEW

REVIEWER	Barakovic, Sabina
REVIEW RETURNED	09-Dec-2020

GENERAL COMMENTS	The study addresses very interesting and contemporary topic: ICTs for the improvement of cognitive function in healthy older adults. A systematic review protocol. It tends to provide as systematic review protocol for ICTs for improvement of cognitive function in healthy older adults by using PRISMA-P methodology. A serious concern is a lack of framework, clear research hypotheses and directions what to do with the results. Also, the review should include engineering and ICT papers as well which are not in the scope of the mentioned databases. The authors should consider the following study as well: https://www.mdpi.com/1660-4601/17/8/2940/htm

REVIEWER	Brustio, Paolo University of Torino, NeuroMuscular Function Research Group, School of Exercise & Sport Sciences , Department of Medical Sciences
REVIEW RETURNED	01-Dec-2020

GENERAL COMMENTS	This protocol aims to plan a systematic review on the effect of cognitive intervention using Information and Communication Technologies in order to improve cognitive function in people aged 55 and over. In my opinion, the authors addressed all the aspects necessary to accurately plan their systematic review. Here my comments to the authors to improve the quality of their work. Page 1 Line 27 Please delete “..” Page 1 Lines 32-35 Citations are needed. Page 5 Line 55

	Please discuss the difference between stimulation and cognitive training interventions. Page 6 lines 12-20 I think that the authors should clarify and deepen more in detail the difference between traditional and alternative cognitive interventions. While the authors focused on the practical aspect of alternative cognitive intervention, no data are reported about the effect and the difference between traditional and alternative cognitive intervention in terms of cognitive function. This evidence may clarify the importance of this systematic metanalysis. Again, what are the difference between the intervention in terms of social factors? I think that difference can arise. Page 6 lines 22-25 Please provide some references. Page 6 line 26-27 The authors explain that the type of interventions, as well as the duration of the intervention, varied. I think that a focus on these aspects can add evidence to the authors' work. Page 6 line 46 Please define the difference between cognitive level and cognitive function or what you intended. Page 7 lines 47-54 In my opinion, it is necessary to clarify the outcomes included in the study. I think that this aspect is important to define what authors intend for cognitive function. Indeed, the authors compared the MMSE or the MOCA with the IQ that evaluated different things. Additionally, also the neurophysiological test evaluates different components of the cognitive function and executive function. I suggest clarifying this aspect. Page 7 line 14 Please better explain the rationality to include participants 55 yrs old or over also in the aim of the study Pages 7-8 You have not even thought about checking the bibliography of eligible articles. In this way, you might find possible articles to be included in your work. Page 9 Lines 43-53 I also suggest discussing the results for participants' characteristics. I think that in this manner your work will increase the quality. Indeed, considering your eligibility criteria the range age of your sample may be larger, and probably more females could be included in the selected studies. I think that the effect of these interventions may change considering age and gender. Please think to add this additional analysis in your future work.

REVIEWER	Almathami, Hassan Khader UOW
REVIEW RETURNED	19-Apr-2021

GENERAL COMMENTS	Line10: Abbreviation should be used after you write the full text “PRESS”

	Line 20: Are you making a claim “although it is too early to make a claim” or someone else is making a claim? “This suggests that the use of ICT ……. to the of the line” if you are not making a claim, please use citation. Line22-27: Paragraph reference!! it looks like it is from reference number 16 “Computerized Cognitive Training in Cognitively Healthy Older Adults: A Systematic Review and Meta-Analysis of Effect Modifiers” Line 48: Abbreviation should be used after you write the full text for “RoB2”. Line50: Would you explain why you won’t include studies that involve people who have diagnosis of cognitive impairment nor accidents or diseases associated with it? Note1: The research question is not very clear. Also, the writing can be improved. Note2: In the whole document, the full stop should be after the citation. Note3: What is the difference between your study and other published systematic reviews? It would be appropriate if you showcase how different your study from other systematic reviews related to your topic. For example, what is the difference between your study and the study titled “Computerized Cognitive Training in Cognitively Healthy Older Adults: A Systematic Review and Meta-Analysis of Effect Modifiers”

VERSION 1 – AUTHOR RESPONSE

Comments from reviewer 1

1. The study addresses very interesting and contemporary topic: ICTs for the improvement of cognitive function in healthy older adults. A systematic review protocol. It tends to provide as systematic review protocol for ICTs for improvement of cognitive function in healthy older adults by using PRISMA-P methodology. A serious concern is a lack of framework, clear research hypotheses and directions what to do with the results. Also, the review should include engineering and ICT papers as well which are not in the scope of the mentioned databases. The authors should consider the following study as well: https://www.mdpi.com/1660-4601/17/8/2940/htm

Thank you very much for your comments to improve the quality of our study. Regarding the comments made:

• We included the IEEE Xplore database. We added this database in the methods section:
“The following databases will be searched, Medline (Pubmed), CinahlPlus, Scopus, Embase, WoS, CENTRAL and IEEE Xplore”

• We very appreciate the suggestion made and consequently included the concept of Quality of Life and mentioned the framework proposed (https://www.mdpi.com/1660-4601/17/8/2940/htm), which has offered us a very useful view on the intended approach.
Changes can be seen along the Introduction section, especially in the beginning and before the ICT solutions:

o “Quality of Life (QoL) have multiple frameworks and according to Brown et al.1 it can be concluded as: “QoL is inherently a dynamic, multi-level and complex concept, reflecting objective, subjective, macro-societal, and micro-individual, positive and negative influences which interact together”. When this concept is applied to elderly the domains ‘Health perception’, ‘Autonomy’, ‘Role and activity’, ‘Attitude and adaption’ and ‘Relationships’ are most covered by the majority of elderly population although other domains like ‘Spirituality’ and ‘Financial security’ are less covered but important by some elderly groups.
o Smart ageing is a term that is often used interchangeably with healthy ageing or active ageing17 but it can be described as intelligently utilizing modern biomedical, digital healthcare, computing, and communication technologies.18 Based on the review of Barakovic et al.17 ICT solutions that are designed for smart ageing contribute to 7 out of 9 smart ageing determinants of its framework: 1) physical activity, 2) new technologies, 3) long-term care, 4) environment and accessibility, 5) social inclusion and participation, 6) diet and nutrition and 7) access to services. For that reason, ICT solutions encompasses a series of technological tools, such as individual computers, video games, internet, applications, etc. which offer an ideal resource for designing both stimulation

• We reformulated the aim of our review in order to make our research question clearer. Also, we linked the objective of the aim to future possible research regarding ICTs and prevention of cognitive impairment, that way we could ascertain the high evidence through this systematic review of RCTs and plan our intervention based on the evidence:

“The aim of this review is to ascertain the scientific evidence on the use of ICT in a healthy adult population or one at risk of dementia for the prevention of cognitive impairment, in order to obtain high quality evidence through this systematic review of RCTs and future interventions could be designed”

Comments from reviewer 2

This protocol aims to plan a systematic review on the effect of cognitive intervention using Information and Communication Technologies in order to improve cognitive function in people aged 55 and over. In my opinion, the authors addressed all the aspects necessary to accurately plan their systematic review. Here my comments to the authors to improve the quality of their work.

1. Page 1 Line 27: Please delete “..”

Thank you for appointing this error. We deleted these double full stops.

2. Page 1 Lines 32-35: Citations are needed.

We added the citations to the phrase : Thus, MCI is considered to be a state of transition between the cognitive alterations characteristic of old age and dementia.10 11
The particular references are:

10. Eshkoor SA, Hamid TA, Mun CY, et al. Mild cognitive impairment and its management in older people. Clin Interv Aging 2015;10:687-93. doi: 10.2147/CIA.S73922
11. Morris JC, Price JL. Pathologic Correlates of Nondemented Aging, Mild Cognitive Impairment, and Early-Stage Alzheimer's Disease. Journal of Molecular Neuroscience 2001;17(2):101-18. doi: 10.1385/jmn:17:2:101

3. Page 5 Line 55 : Please discuss the difference between stimulation and cognitive training interventions.

Thank you very much for your comment, we added a clarification and discussed the differences between stimulation and cognitive training interventions in the introduction section:

“For that reason, ICT solutions encompasses a series of technological tools, such as individual computers, video games, internet, applications, etc. which offer an ideal resource for designing both stimulation interventions, designed to participate in activities to improve cognitive and social functioning,19 and cognitive training interventions, designed as a guided standardized tasks to enhance particular cognitive functions.”19

4. Page 6 lines 12-20: I think that the authors should clarify and deepen more in detail the difference between traditional and alternative cognitive interventions. While the authors focused on the practical aspect of alternative cognitive intervention, no data are reported about the effect and the difference between traditional and alternative cognitive intervention in terms of cognitive function. This evidence may clarify the importance of this systematic metanalysis. Again, what are the difference between the intervention in terms of social factors? I think that difference can arise.

We highly agree with the reviewer on adding this topic. We included this aspect and highlighted the advantages of the interventions using ICTs by adding results from other already carried out studies.

“The results found on the use of ICTs for the prevention of cognitive impairment seem to have similar or even better effects compared to more traditional cognitive training approaches, where a trained person is required to carry out the face-to-face sessions, with the consequent needs for travel, physical space and timetables among other requirements.28 29 Those beneficial outcomes on CCT are shown in different studies, a prior systematic review (SR) with meta-analysis evidenced that CCT is effective at enhancing cognitive function in healthy older adults specifically for nonverbal memory, processing speed, working memory, and visuospatial outcomes, but not for attention and executive functions.23 A Randomized Controlled Trial (RCT) showed improvements in cognitively unimpaired older adults with multiple dementia risks factors on memory and processing speed, and on global cognition.30 And a recent RCT demonstrates that CCT provides benefits on executive functions, but when the CCT is combined with exercise it improved multiple cognitive processes of executive functions.31 The type of interventions have varied as well as the duration of these said interventions, the SR of Lampit and cols.23 evidenced that training sessions undertaken more than three times a week or sessions lasting less than 30 minutes may be ineffective. Moreover, the use of CCT can be directed to a specific cognitive function, it can be personalized to individuals, the use of gamification led to a more enjoyable and immersive experience, it can provide quantitative feedback immediately and it can be delivered on common and portable digital plaforms.32 “

Moreover we consider that a specification in CCS was needed, therefore we added a paragraph defining this term:

On the other hand, the use of technologies to achieve a general enhancement of cognitive and social functioning is known as Computerized Cognitive Stimulation (CCS)24. CCS programs demonstrates

that can improve episodic memory.25

5. Page 6 lines 22-25: Please provide some references.

We added a reference to support our statement:

“As a result, there has been an exponential increase in research in the last ten years using different types of ICTs to prevent dementia in the healthy adult population or those at risk of cognitive deterioration.33”

33. Ten Brinke LF, Davis JC, Barha CK, et al. Effects of computerized cognitive training on neuroimaging outcomes in older adults: a systematic review. BMC Geriatr 2017;17(1):139. doi: 10.1186/s12877-017-0529-x

6. Page 6 line 26-27: The authors explain that the type of interventions, as well as the duration of the intervention, varied. I think that a focus on these aspects can add evidence to the authors' work.

Thank you very much for your comment, we redistributed this paragraph according to the last comment, and added references to this paragraph:

“The type of interventions has varied as well as the duration of these said interventions, the SR of Lampit and cols.23 evidenced that training sessions undertaken more than three times a week or sessions lasting less than 30 minutes may be ineffective.”

7. Page 6 line 46: Please define the difference between cognitive level and cognitive function or what you intended.

Thank you very much for your comment, we deleted the cognitive level approach taking into account that we will focus our review on the improvement of cognitive functions:

RCTs will be included in all of the designs that use ICTs to improve cognitive functions to prevent MCI. Information obtained from books, master's or doctoral theses, conference papers, reports or scientific posters will be excluded.

8. Page 7 lines 47-54: In my opinion, it is necessary to clarify the outcomes included in the study. I think that this aspect is important to define what authors intend for cognitive function. Indeed, the authors compared the MMSE or the MOCA with the IQ that evaluated different things. Additionally, also the neurophysiological test evaluates different components of the cognitive function and executive function. I suggest clarifying this aspect.

We clarified these aspects and this section has been rewritten:

“Articles will be included that report data on global cognitive function measured through validated instruments such as the Montreal Cognitive Assessment (MOCA)41 or the Minimental-State Examination (MMSE)42. We will also considerate cognitive tests that measures specific cognitive

function subdomains such as verbal memory, nonverbal memory, working memory (WM), processing speed, attention, language, visuospatial skills, and executive functions."

9. Page 7 line 14: Please better explain the rationality to include participants 55 yrs old or over also in the aim of the study

Thank you very much for your comment, we followed your recommendations and we broaden our sample to people of 50 years old and over, as we noticed recent research and interventions related to smart ageing starts at this age.

10. Pages 7-8: You have not even thought about checking the bibliography of eligible articles. In this way, you might find possible articles to be included in your work.

We clarified this point and added bibliography of eligible studies In our search method:

"The following databases will be searched, Medline (Pubmed), CinahlPlus, Scopus, Embase, WoS, CENTRAL and IEEE Xplore, the search will be completed on ClinicalTrials.gov and Google Scholar, as well as a manual search, checking the bibliography of the eligible studies and the indexes of relevant journals, to ensure we do not lose any results."

11. Page 9 Lines 43-53: I also suggest discussing the results for participants' characteristics. I think that in this manner your work will increase the quality. Indeed, considering your eligibility criteria the range age of your sample may be larger, and probably more females could be included in the selected studies. I think that the effect of these interventions may change considering age and gender. Please think to add this additional analysis in your future work.

We strongly agree with the reviewer and think discussing these characteristics could contribute in the quality of the interpretation of the results. We added these topics and rewrote this section

"A narrative synthesis of the results will be made, according to the objectives of each intervention and according to the characteristics described in the data extraction. The data will be presented according to the technology used for the improvement of the cognitive function taking into account age and gender of participants"

Comments from reviewer 3

1. Line10: Abbreviation should be used after you write the full text "PRESS"

Thank you very much for your comment, we added the full text:

"The search strategy for Medline (Pubmed) will be designed (see annex 1) and will follow the recommendations of the Peer Review of Electronic Search Strategies (PRESS) Statement."

2. Line 20: Are you making a claim "although it is too early to make a claim" or someone else is making a claim? "This suggests that the use of ICT …… to the of the line" if you are not making a claim, please use citation.

We clarified that this statement was made in the already published literature and we added the citation supporting this claim:

“This suggests that the use of ICT is a promising preventive alternative for cognitive impairment and dementia.33-35”

33. Ten Brinke LF, Davis JC, Barha CK, et al. Effects of computerized cognitive training on neuroimaging outcomes in older adults: a systematic review. BMC Geriatr 2017;17(1):139. doi: 10.1186/s12877-017-0529-x
34. Rebok GW, Ball K, Guey LT, et al. Ten-year effects of the advanced cognitive training for independent and vital elderly cognitive training trial on cognition and everyday functioning in older adults. J Am Geriatr Soc 2014;62(1):16-24. doi: 10.1111/jgs.12607
35. Ten Brinke LF, Best JR, Crockett RA, et al. The effects of an 8-week computerized cognitive training program in older adults: a study protocol for a randomized controlled trial. BMC Geriatr 2018;18(1):31. doi: 10.1186/s12877-018-0730-6

3. Line22-27: Paragraph reference!! it looks like it is from reference number 16 “Computerized Cognitive Training in Cognitively Healthy Older Adults: A Systematic Review and Meta-Analysis of Effect Modifiers”

We agree with the reviewer that a reference for this paragraph was missing. We added the used reference:

“As a result, there has been an exponential increase in research in the last ten years using different types of ICTs to prevent dementia in the healthy adult population or those at risk of cognitive deterioration.33”

33. Ten Brinke LF, Davis JC, Barha CK, et al. Effects of computerized cognitive training on neuroimaging outcomes in older adults: a systematic review. BMC Geriatr 2017;17(1):139. doi: 10.1186/s12877-017-0529-x

4. Line 48: Abbreviation should be used after you write the full text for “RoB2”.

Thank you very much for your comment, we added the full text of RoB2

“Two independent researchers will analyse the quality assessment according to the revised tool for assessing risk of bias in randomised trials(RoB 2).46”

5. Line50: Would you explain why you won’t include studies that involve people who have diagnosis of cognitive impairment nor accidents or diseases associated with it?

Thank you very much for your comment, we have write again the objective of our study according to this comment and the next one.

The main objective of the systematic review will be to review and analyse the use of ICTs for the improvement of cognitive functions in healthy older adult population aged 50 and over, for the prevention of cognitive impairment not related to accidents or diseases associated with it.

6. Note1: The research question is not very clear. Also, the writing can be improved.

Thank you very much for your comment, we clarified our research question and stated:

The main objective of the systematic review will be to review and analyse the use of ICTs for the improvement of cognitive functions in healthy older adult population aged 50 and over, for the prevention of cognitive impairment not related to accidents or diseases associated with it.

7. Note2: In the whole document, the full stop should be after the citation.

Thank you for your comment, we followed the instructions of the BMJ Author Hub and used the modified Vancouver Style available in EndNote and where authors are asked to insert references numbers immediately after punctuation (with no word spacing). In the BMJ Author Hub, reference numbers are in brackets, however, the endnote style is in super index. We would like to ask the reviewer or editorial team indicate us the adequate reference style in order to make the appropriate changes.

https://authors.bmj.com/writing-and-formatting/formatting-your-paper/#references

8. Note3: What is the difference between your study and other published systematic reviews? It would be appropriate if you showcase how different your study from other systematic reviews related to your topic. For example, what is the difference between your study and the study titled “Computerized Cognitive Training in Cognitively Healthy Older Adults: A Systematic Review and Meta-Analysis of Effect Modifiers”

Our systematic review will include all RCTs to improve cognitive functions for cognitive stimulation or cognitive training to prevent Cognitive Impairment in healthy adult population. In consequence, we will not exclude RCTs by the duration of the trial, this point could be a limitation in our systematic review. The difference of our systematic review with the one mentioned is.
• Date of the published SR. Lampit et al. published this SR in 2014, we think that more ICT solutions will be available since the publication of this SR
• We added more databases and also IEEE Xplore as technological database. For that reason, the searches and the retrieval of studies will be more comprehensive.

We also have written again the strengths and limitations section to showcase the strengths and limitations of our SR

• The systematic review will follow the recommendations of the Peer Review of Electronic Search Strategies (PRESS) Statement
• The systematic review will include adult population over 50 years with randomized controlled trials (RCTs) that are targeted by Computerized Cognitive Training (CCT) and/or Computerized Cognitive Stimulation (CCS).
• All RCTs that meet the inclusion and exclusion criteria will be included regardless of the duration of the intervention and its follow-up to obtain a broader insight of the current evidence.
• Only studies written in English or Spanish will be included, that could cause a language bias.
• A comprehensive search in health, multidisciplinary and technological databases will be conducted.

VERSION 2 – REVIEW

REVIEWER	Almathami, Hassan Khader

	UOW
REVIEW RETURNED	09-Aug-2021

GENERAL COMMENTS	The authors addressed all my comments adequately. I would only suggest proofreading the paper before publishing it.